

ANSIBLE

Tasks To Be Performed:

1. Setup Ansible cluster with 3 nodes

```
[ec2-user@ip-172-31-44-130 etc]$ ansible localhost -m ping
```

```
[WARNING]: No inventory was parsed, only implicit localhost is available
```

```
localhost | SUCCESS => {
```

```
    "changed": false,
```

```
    "ping": "pong"
```

```
}
```

```
[ec2-user@ip-172-31-44-130 etc]$
```

Passwordless ssh auth is set up from master to below nodes.

Target servers are updated under /etc/ansible/hosts

[webservers]

172.31.36.67

[dbservers]

172.31.38.158

```
[ec2-user@ip-172-31-44-130 ansible]$ ansible webservers -m ping
```

```
[WARNING]: Platform linux on host 172.31.36.67 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of
```

```
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
```

```
172.31.36.67 | SUCCESS => {
```

```
    "ansible_facts": {
```

```
        "discovered_interpreter_python": "/usr/bin/python3.9"
```

```
    },
```

```
    "changed": false,
```

```
    "ping": "pong"
```

```
}
```

```
[ec2-user@ip-172-31-44-130 ansible]$ ansible dbservers -m ping
```

```
[WARNING]: Platform linux on host 172.31.38.158 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation
```

```
of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
```

```
172.31.38.158 | SUCCESS => {
```

```
    "ansible_facts": {
```

```
        "discovered_interpreter_python": "/usr/bin/python3.9"
```

```
    },
```

```
"changed": false,  
"ping": "pong"  
}  
[ec2-user@ip-172-31-44-130 ansible]$
```

2. On slave 1 install Java

3. On slave 2 install MySQL server

Do the above tasks using Ansible Playbooks

```
[ec2-user@ip-172-31-44-130 ansible]$ cat first_playbook.yaml
```

```
---
```

```
- name: Install java on slave1  
  hosts: webserver  
  become: yes  
  tasks:  
    - name: Install java  
      ansible.builtin.dnf:  
        name: java-17-amazon-corretto  
        state: present  
- name: Install MySQL on slave2  
  hosts: dbserver  
  become: yes
```

```
tasks:  
  - name: Install mysql  
    ansible.builtin.dnf:  
      name: mariadb118-server  
      state: present  
  - name: Start MySQL service  
    ansible.builtin.service:  
      name: mariadb118-server  
      state: started  
      enabled: yes
```

```
[ec2-user@ip-172-31-44-130 ansible]$ sudo vi first_playbook.yaml
```

```
[ec2-user@ip-172-31-44-130 ansible]$ ansible-playbook first_playbook.yaml --syntax-check
```

```
playbook: first_playbook.yaml
```

```
[ec2-user@ip-172-31-44-130 ansible]$ ansible-playbook first_playbook.yaml
```

```
-----
```

PLAY [Install java on slave1]

TASK [Gathering Facts]

[WARNING]: Platform linux on host 172.31.36.67 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of

another Python interpreter could change the meaning of that path. See <https://docs.ansible.com/ansible->

core/2.15/reference_appendices/interpreter_discovery.html for more information.

ok: [172.31.36.67]

TASK [Install java]

ok: [172.31.36.67]

PLAY [Install MySQL on slave2]

TASK [Gathering Facts]

[WARNING]: Platform linux on host 172.31.38.158 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation

of another Python interpreter could change the meaning of that path. See <https://docs.ansible.com/ansible->

core/2.15/reference_appendices/interpreter_discovery.html for more information.

ok: [172.31.38.158]

TASK [Install mysql]

ok: [172.31.38.158]

TASK [Start MySQL service]

changed: [172.31.38.158]

PLAY RECAP

172.31.36.67 : ok=2 changed=0 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0

```
172.31.38.158      : ok=3  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
```

```
[ec2-user@ip-172-31-44-130 ansible]$
```

Tasks To Be Performed:

1. Create a script which can add text “This text has been added by custom script” to /tmp.1.txt

This file name was considered as /tmp/1.txt instead of /tmp.1.txt.

```
[ec2-user@ip-172-31-44-130 ansible]$ cat custom_script.sh
```

```
#!/bin/sh
```

```
echo "This text has been added by custom script" > /tmp/1.txt
```

2. Run this script using Ansible on all the hosts

```
[ec2-user@ip-172-31-44-130 ansible]$ cat second_playbook.yaml
```

```
---
```

```
- hosts: all
```

```
  become: true
```

```
  name: play1
```

```
  tasks:
```

```
    - name: Script to add text
```

```
      ansible.builtin.script: custom_script.sh
```

```
[ec2-user@ip-172-31-44-130 ansible]$
```

```
[ec2-user@ip-172-31-44-130 ansible]$ ansible-playbook second_playbook.yaml --syntax-check
```

```
playbook: second_playbook.yaml
```

```
[ec2-user@ip-172-31-44-130 ansible]$ ansible-playbook second_playbook.yaml
```

```
PLAY [play1]
```

```
*****
*****
```

```
TASK [Gathering Facts]
```

```
*****
***
```

```
[WARNING]: Platform linux on host 172.31.36.67 is using the discovered Python interpreter at /usr/bin/python3.9,
but future installation of
```

```
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.15/reference_appendices/interpreter_discovery.html for more information.
```

```
ok: [172.31.36.67]
```

```
[WARNING]: Platform linux on host 172.31.38.158 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation
```

```
of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
```

core/2.15/reference_appendices/interpreter_discovery.html for more information.

ok: [172.31.38.158]

TASK [Script to add text]

changed: [172.31.36.67]

changed: [172.31.38.158]

PLAY RECAP

172.31.36.67 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

172.31.38.158 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

[ec2-user@ip-172-31-44-130 ansible]\$

Tasks To Be Performed:

1. Create 2 Ansible roles

2. Install Apache2 on slave1 using one role and NGINX on slave2 using the other role

Slave 1:

ubuntu@ip-172-31-32-34:~/.ssh\$ **systemctl status apache2**

Unit apache2.service could not be found.

ubuntu@ip-172-31-32-34:~/.ssh\$ **systemctl status apache2**

● apache2.service - The Apache HTTP Server

Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)

Active: active (running) since Mon 2026-06-22 09:16:08 UTC; 15s ago

Invocation: 5aaf56e46ee1492687161f86abd50997

Docs: <https://httpd.apache.org/docs/2.4/>

Main PID: 3464 (apache2)

Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"

Tasks: 55 (limit: 627)

Memory: 5.9M (peak: 5.9M)

CPU: 127ms

CGroup: /system.slice/apache2.service

|—3464 /usr/sbin/apache2 -k start -DFOREGROUND

|—3480 /usr/sbin/apache2 -k start -DFOREGROUND

|—3482 /usr/sbin/apache2 -k start -DFOREGROUND

Jun 22 09:16:08 ip-172-31-32-34 systemd[1]: Starting apache2.service - The Apache HTTP Server...

Jun 22 09:16:08 ip-172-31-32-34 systemd[1]: Started apache2.service - The Apache HTTP Server.

ubuntu@ip-172-31-32-34:~/.ssh\$

Slave2:

[ec2-user@ip-172-31-38-158 ~]\$ systemctl status nginx

● nginx.service - The nginx HTTP and reverse proxy server

Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)

Active: active (running) since Mon 2026-06-22 09:32:01 UTC; 26s ago

Process: 8887 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)

Process: 8888 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)

Process: 8889 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)

Main PID: 8890 (nginx)

Tasks: 3 (limit: 1014)

Memory: 3.4M

CPU: 63ms

CGroup: /system.slice/nginx.service

├─8890 "nginx: master process /usr/sbin/nginx"

├─8891 "nginx: worker process"

└─8892 "nginx: worker process"

Jun 22 09:32:01 ip-172-31-38-158.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...

Jun 22 09:32:01 ip-172-31-38-158.ap-south-1.compute.internal nginx[8888]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok

Jun 22 09:32:01 ip-172-31-38-158.ap-south-1.compute.internal nginx[8888]: nginx: configuration file /etc/nginx/nginx.conf test is successful

Jun 22 09:32:01 ip-172-31-38-158.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.

[ec2-user@ip-172-31-38-158 ~]\$

3. Above should be implemented using different Ansible roles

[ec2-user@ip-172-31-44-130 ansible_roles_project]\$ tree

```
.
├── apache
│   ├── README.md
│   ├── defaults
│   │   └── main.yml
│   ├── files
│   └── handlers
```

```

| | └── main.yml
| └── meta
| | └── main.yml
| └── tasks
| | └── main.yml
| └── templates
| └── tests
| | └── inventory
| | └── test.yml
| └── vars
| └── main.yml
└── inventory.ini
└── nginx
| └── README.md
| └── defaults
| | └── main.yml
| └── files
| └── handlers
| | └── main.yml
| └── meta
| | └── main.yml
| └── tasks
| | └── main.yml
| └── templates
| └── tests
| | └── inventory
| | └── test.yml
| └── vars
| └── main.yml
└── site.yml

```

18 directories, 18 files

[ec2-user@ip-172-31-44-130 ansible_roles_project]\$

[ec2-user@ip-172-31-44-130 ansible_roles_project]\$ ll

total 8

drwxr-xr-x. 10 root root 135 Jun 22 08:51 apache

-rw-r--r--. 1 root root 87 Jun 22 09:21 inventory.ini

```
drwxr-xr-x. 10 root root 135 Jun 22 09:20 nginx
-rw-r--r--.  1 root root 111 Jun 22 09:23 site.yml
[ec2-user@ip-172-31-44-130 ansible_roles_project]$
```

```
[ec2-user@ip-172-31-44-130 ansible_roles_project]$ cat inventory.ini
```

```
[apache]
```

```
172.31.32.34 ansible_user=ubuntu
```

```
[nginx]
```

```
172.31.38.158 ansible_user=ec2-user
```

```
[ec2-user@ip-172-31-44-130 ansible_roles_project]$
```

```
[ec2-user@ip-172-31-44-130 ansible_roles_project]$ cat site.yml
```

```
---
```

```
- hosts: apache
```

```
  become: true
```

```
  roles:
```

```
    - apache
```

```
- hosts: nginx
```

```
  become: true
```

```
  roles:
```

```
    - nginx
```

```
[ec2-user@ip-172-31-44-130 tasks]$ pwd
```

```
/etc/ansible/ansible_roles_project/apache/tasks
```

```
[ec2-user@ip-172-31-44-130 tasks]$ cat main.yml
```

```
---
```

```
# tasks file for apache
```

```
- name: Install Apache2
```

```
  ansible.builtin.apt:
```

```
    name: apache2
```

```
    state: present
```

```
    update_cache: yes
```

```
- name: Start Apache service
```



```
ansible.builtin.service:
```

```
  name: apache2
```

```
  state: started
```

```
  enabled: yes
```

```
[ec2-user@ip-172-31-44-130 tasks]$ pwd
```

```
/etc/ansible/ansible_roles_project/nginx/tasks
```

```
[ec2-user@ip-172-31-44-130 tasks]$ cat main.yml
```

```
---
```

```
# tasks file for nginx
```

```
- name: Install nginx
```

```
  ansible.builtin.yum:
```

```
    name: nginx
```

```
    state: present
```

```
    update_cache: yes
```

```
- name: Start service
```

```
  ansible.builtin.service:
```

```
    name: nginx
```

```
    state: started
```

```
    enabled: true
```

```
[ec2-user@ip-172-31-44-130 tasks]$
```

Tasks To Be Performed:

1. Use the previous deployment of Ansible cluster

2. Configure the files folder in the role with index.html which should be replaced with the original index.html

Created index.html under **/etc/ansible/ansible_roles_project/nginx/files**

```
<html>
```

```
<h1> “This page is rendered from Slave1 server - Nginx host”</h1>
```

```
</html>
```

```
[ec2-user@ip-172-31-44-130 tasks]$ pwd
```

```
/etc/ansible/ansible_roles_project/nginx/tasks
```

```
[ec2-user@ip-172-31-44-130 tasks]$ cat main.yml
```

```
---
```

```
# tasks file for nginx
```

```
- name: Install nginx
```

```
  ansible.builtin.yum:
```

```
name: nginx
state: present
update_cache: yes
```

```
- name: Start service
  ansible.builtin.service:
    name: nginx
    state: started
    enabled: true
```

- name: Copy custom html page

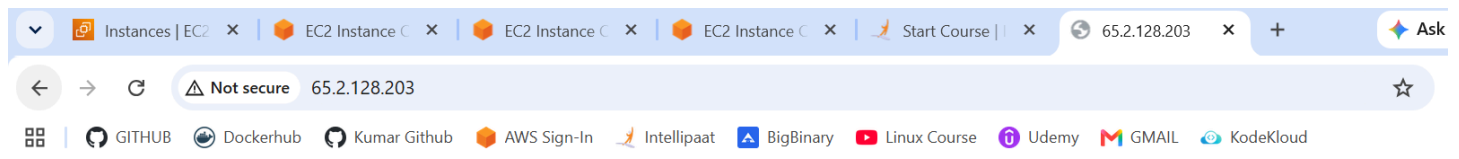
ansible.builtin.copy:

src: index.html

dest: /usr/share/nginx/html/index.html

[ec2-user@ip-172-31-44-130 tasks]\$

All of the above should only happen on the slave which has NGINX installed using the role



"This page is rendered from Slave1 server - Nginx host"

Tasks To Be Performed:

1. Create a new deployment of Ansible cluster of 5 nodes

Instances (5) [Info](#) Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets: Running

Instance state = running [X](#) [Clear filters](#)

☐	Name ✎	Instance ID	Instance state	Instance type	Status
☐	Master_Node	i-034b92d94609b9d42	Running	t3.micro	3/3
☐	Test_1	i-02cb36aee8ec0baab	Running	t3.micro	3/3
☐	Test_2	i-020209e73fb7876f9	Running	t3.micro	3/3
☐	Prod_1	i-01682f84926168032	Running	t3.micro	3/3
☐	Prod_2	i-0d56c78417a2b4aeb	Running	t3.micro	3/3

2. Label 2 nodes as test and other 2 as prod

3. Install Java on test nodes

Test1:

```
[ec2-user@ip-172-31-38-211 .ssh]$ java
```

```
-bash: java: command not found
```

```
[ec2-user@ip-172-31-38-211 .ssh]$ java
```

```
Usage: java [java options...] <application> [application arguments...]
```

Where <application> is one of:

- <mainclass> to execute the main method of a compiled main class
- jar <jarfile>.jar to execute the main class of a JAR archive
- m <module>[/<mainclass>] to execute the main class of a module
- <sourcefile>.java to compile and execute a source-file program

Test2:

```
[ec2-user@ip-172-31-34-68 .ssh]$ java
```

```
-bash: java: command not found
```

```
[ec2-user@ip-172-31-34-68 .ssh]$ java
```

```
Usage: java [java options...] <application> [application arguments...]
```

Where <application> is one of:

- <mainclass> to execute the main method of a compiled main class
- jar <jarfile>.jar to execute the main class of a JAR archive
- m <module>[/<mainclass>] to execute the main class of a module
- <sourcefile>.java to compile and execute a source-file program

Where key java options include:

--class-path <class path>

where <class path> is a list of directories and JAR archives to search for class files, separated by ":"

--module-path <module path>

where <module path> is a list of directories and JAR archives to search for modules, separated by ":"

-version

to print product version to the error stream and exit

For additional help on usage: `java --help`

For an interactive Java environment: `jshell`

[ec2-user@ip-172-31-34-68 .ssh]\$

4. Install MySQL server on prod nodes

Prod1:

[ec2-user@ip-172-31-41-119 .ssh]\$ systemctl status mariadb

● mariadb.service - MariaDB 11.8 database server

Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)

Active: active (running) since Tue 2026-06-23 08:51:19 UTC; 50s ago

Docs: man:mariabdb(8)

<https://mariadb.com/kb/en/library/systemd/>

Process: 36551 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)

Process: 36674 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)

Process: 36928 ExecStartPre=/bin/sh -c [! -e /usr/bin/galera_recovery] && VAR= || VAR=/usr/bin/galera_recovery; [\$? -eq 0] &&>

Process: 36943 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)

Process: 36973 ExecStartPost=/bin/rm -f /run/mariadb/wsrep-start-position (code=exited, status=0/SUCCESS)

Main PID: 36929 (mariabdb)

Status: "Taking your SQL requests now..."

Tasks: 13 (limit: 1059)

Memory: 96.2M

CPU: 1.539s

CGroup: /system.slice/mariadb.service

└─36929 /usr/libexec/mariabdb --basedir=/usr

Prod2:

[ec2-user@ip-172-31-37-59 ~]\$ systemctl status mariadb

● mariadb.service - MariaDB 11.8 database server

Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)

Active: active (running) since Tue 2026-06-23 08:51:19 UTC; 36s ago

Docs: man:mariadb(8)

<https://mariadb.com/kb/en/library/systemd/>

Process: 36505 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)

Process: 36615 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)

Process: 36902 ExecStartPre=/bin/sh -c [! -e /usr/bin/galera_recovery] && VAR= || VAR=/usr/bin/galera_recovery; [\$? -eq 0] &&>

Process: 36916 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)

Process: 36946 ExecStartPost=/bin/rm -f /run/mariadb/wsrep-start-position (code=exited, status=0/SUCCESS)

Main PID: 36903 (mariabdb)

Use Ansible roles for the above and group the hosts under test and prod.

[ec2-user@ip-172-31-45-177 ansible_project]\$ tree

```
.
├── inventory.ini
├── java
│   ├── README.md
│   ├── defaults
│   │   └── main.yml
│   ├── files
│   ├── handlers
│   │   └── main.yml
│   ├── meta
│   │   └── main.yml
│   ├── tasks
│   │   └── main.yml
│   ├── templates
│   ├── tests
│   │   ├── inventory
│   │   └── test.yml
│   └── vars
│       └── main.yml
├── mysql
│   ├── README.md
│   ├── defaults
│   │   └── main.yml
│   └── files
```

```
|   ├── handlers
|   |   └── main.yml
|   ├── meta
|   |   └── main.yml
|   ├── tasks
|   |   └── main.yml
|   ├── templates
|   ├── tests
|   |   ├── inventory
|   |   └── test.yml
|   └── vars
|       └── main.yml
└── site.yml
```

18 directories, 18 files

```
[ec2-user@ip-172-31-45-177 ansible_project]$ ll
```

total 8

```
-rw-r--r--. 1 root root 69 Jun 23 08:03 inventory.ini
```

```
drwxr-xr-x. 10 root root 135 Jun 23 07:59 java
```

```
drwxr-xr-x. 10 root root 135 Jun 23 07:59 mysql
```

```
-rw-r--r--. 1 root root 101 Jun 23 08:30 site.yml
```

```
[ec2-user@ip-172-31-45-177 ansible_project]$ cat java/tasks/main.yml
```

```
# tasks file for java
```

```
- name: Java installation
```

```
  ansible.builtin.yum:
```

```
    name: java-26-amazon-corretto.x86_64
```

```
    state: present
```

```
    update_cache: yes
```

```
[ec2-user@ip-172-31-45-177 ansible_project]$ cat mysql/tasks/main.yml
```

```
# tasks file for mysql
```

```
- name: Install mysql on prod nodes
```

```
  ansible.builtin.yum:
```

```
    name: mariadb118-server
```

state: present

- name: Start service

ansible.builtin.service:

name: mariadb

state: started

enabled: yes

[ec2-user@ip-172-31-45-177 ansible_project]\$